



**ARULMIGU PALANIANDAVAR COLLEGE OF
ARTS AND CULTURE**

PALANI – 624601

PROJECT REPORT

ON

**CROP YIELD PREDICTION
USING MACHINE LEARNING**

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1. ABOUT THE PROJECT

1.1 ABSTRACT

India is an agricultural nation and depends a lot on its agricultural activities. Prediction of crop yield is a necessary activity that allows farmers to estimate storage, optimize resources, increases efficiency and decrease costs. However, farmers usually predict crops based on the region, soil, weather conditions and the crop itself based on experience and estimates which may not be very accurate especially with the constantly changing and unpredictable climatic conditions of the present day. To solve this problem, we aim to predict the yield of various crops such as Rice, maize etc., According to our analysis, the most used features are temperature, rainfall and soil type. Numerous studies have been conducted in the agricultural sector to better estimate crop yield through machine learning techniques. It is an efficient method for anticipating crop yields and determining which crops to cultivate. Random Forest has been widely utilized for this purpose. A set of parameters in the Random Forest classifier must be stay tuned. The machine learning algorithm will yield better results with correct hyper parameter adjustment.

Machine Learning (ML) is the best technique which gives a better practical solution to crop yield problem. So the Random Forest algorithm which we decided to used to train our model to give high accuracy and best prediction.. The predicted accuracy of the model is analyzed 95% to 99%. This system will assist farmers in deciding which crop to plant in order to maximise production. This study offers a quick examination of agricultural yield forecasting using the Random Forest (RF) technique. The purpose of this study is to apply the crop yield prediction approach into action to assist farmers in solving agricultural production concerns.

1.2 LIST OF FIGURES

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1.3 LIST OF ABBREVIATIONS

ML	-	Machine Learning
RF	-	Random Forest
AI	-	Artificial Intelligence

2. INTRODUCTION

2.INTRODUCTION

Agriculture is the engine that drives the Indian economy. Agricultural yield is essential in ensuring global food security since it influences the amount of food that can be produced for consumption. The demand for food increases in direct proportion to the growth in world population. Improving agricultural production is thus essential for catering to this demand, alleviating hunger, and ensuring a sustainable food supply. The technology can analyze large datasets and find patterns to predict crop yields with a high degree of accuracy.

Applying ML to agriculture may result in sustainable farming methods. Farmers can use less water, fertilizer, and pesticides by using precise crop production projections, which can contribute to environmental protection. Agricultural forecasting, commonly referred to as crop yield prediction using ML, is the practice of forecasting crop yields using machine learning algorithms. This technology can assist farmers and agricultural specialists in making more educated decisions about crop planting, harvesting, and management.

Machine learning is a practical approach that can deliver more accurate yield prediction based on a variety of parameters. It is capable of learning from datasets by discovering correlations and patterns. Datasets that reflect the outcomes of prior experiments must be used to train the models . Predictive models make predictions based on historical information. Various machine-learning techniques have been explored to improve agricultural yield Prediction. Crop yield prediction using ML techniques typically relies on a variety of data inputs, including weather, soil, previous yield, and other environmental factors. Machine learning approaches for predicting crops analyze these data sources to detect patterns and forecast future crop yields.

This study primarily concentrates on the Random Forest which is applied to predict the yield of crops, which aids farmers in choosing and cultivating the most lucrative crop, so lowering the likelihood of loss and raising productivity and the value of farming land.

2.1 MACHINE LEARNING

Machine learning is a subset of artificial intelligence (AI) that enables a system to autonomously learn and improve using neural networks and deep learning, without being explicitly programmed, by feeding it large amounts of data. Machine learning allows computer systems to continuously adjust and enhance themselves as they accrue more “experiences.” Thus, the performance of these systems can be improved by providing larger and more varied datasets to be processed.

Machine learning (ML) approaches are used in many fields, ranging from supermarkets to evaluate the behavior of customers to the prediction of customers’ phone use. Machine learning is also being used in agriculture for several years. Crop yield prediction is one of the challenging problems in precision agriculture, and many models have been proposed and validated so far. This problem requires the use of several datasets since crop yield depends on many different factors such as climate, weather, soil, use of fertilizer, and seed variety. This indicates that crop yield prediction is not a trivial task; instead, it consists of several complicated steps. Nowadays, crop yield prediction models can estimate the actual yield reasonably, but a better performance in yield prediction is still desirable.

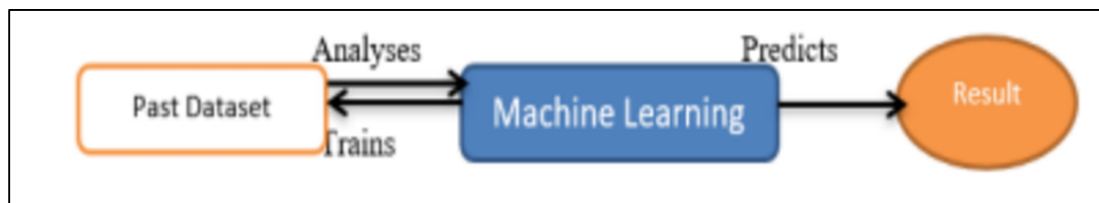


Fig.1: Process of Machine Learning

Machine learning techniques are being widely adopted in agriculture to monitor and predict environmental factors. These techniques build prediction models using data and have applications across various domains. However, developing accurate and interpretable models in crop production prediction remains challenging due to multifaceted influences. Various methods, including field surveys, statistical models, crop growth models, and remote sensing, have been employed in crop yield prediction.

2.2 MACHINE LEARNING APPLICATION

Machine learning is one of the most exciting technologies that one would have ever come across. As is evident from the name, it gives the computer that which makes it more similar to humans: The ability to learn. Machine learning is actively being used today, perhaps in many more places than one would expect.

Today, companies are using Machine Learning to improve business decisions, increase productivity, detect disease, forecast weather, and do many more things. With the exponential growth of technology, we not only need better tools to understand the data we currently have, but we also need to prepare ourselves for the data we will have. To achieve this goal we need to build intelligent machines. We can write a program to do simple things.

But most of the time, Hardwiring Intelligence in it is difficult. The best way to do it is to have some way for machines to learn things themselves. A mechanism for learning – if a machine can learn from input then it does the hard work for us. This is where Machine Learning comes into action. Below are some most trending real-world applications of Machine Learning:

- Image Recognition
- Crop Prediction
- Speech Recognition
- Recommender Systems
- Fraud Detection
- Self Driving Cars
- Medical Diagnosis
- Stock Market Trading
- Virtual Try On

One of the most important uses of machine learning in agriculture is the prediction of crop output. To forecast the anticipated crop yield, historical and current data are used. Farmers and other agricultural industry participants may make well-informed decisions about resource allocation, production planning, and risk management by utilizing machine learning algorithms.

2.3 TYPES OF MACHINE LEARNING

Machine learning is a subset of AI, which enables the machine to automatically learn from data, improve performance from past experiences, and make predictions. Machine learning contains a set of algorithms that work on a huge amount of data. Data is fed to these algorithms to train them, and on the basis of training, they build the model & perform a specific task.

Based on the methods and way of learning, machine learning is divided into mainly four types, which are:

- Supervised Machine Learning
- Unsupervised Machine Learning
- Semi-Supervised Machine Learning
- Reinforcement Learning

1. Supervised machine learning:

It is a type of machine learning where the model is trained on a labeled dataset (i.e., the target or outcome variable is known). For instance, if data scientists were building a model for tornado forecasting, the input variables might include date, location, temperature, wind flow patterns and more, and the output would be the actual tornado activity recorded for those days. Supervised learning is commonly used for risk assessment, image recognition, predictive analytics and fraud detection, and comprises several types of algorithms.

- Regression algorithms
- Classification algorithms
- Naïve Bayes classifiers
- Neural networks
- Random Forest Algorithms

Categories of supervised machine learning

Supervised machine learning can be classified into two type of problems, which are given below:

- Classification
- Regression

Classification: The classification algorithms predict the categories present in the dataset. Some popular classification algorithms are given below:

- Random Forest Algorithm
- Decision Tree Algorithm
- Logistic Regression Algorithm
- Support vector machine Algorithm

Regression: Regression algorithms are used to solve regression problems in which there is a linear relationship between input and output variable.

2. Unsupervised machine learning:

Unsupervised learning is a type of machine learning technique in which an algorithm discovers patterns and relationships using unlabeled data. Unlike supervised learning, unsupervised learning doesn't involve providing the algorithm with labeled target outputs. The primary goal of unsupervised learning is often to discover hidden pattern, similarities, or clusters with in data, within the data, which can then be used for various purpose, such as data exploration, visualization, dimensionality reduction, and more.

3.Semi-supervised learning:

Semi- supervised learning is a Machine learning algorithm that works between the supervised and unsupervised learning so it uses both labeled and unlabeled data. Semi-supervised learning is chosen when labeled data requires skills and relevant resources in order to train or learn from it. We can use the unsupervised techniques to predict labels and then feed these labels to supervised techniques. This techniques is mostly applicable in the case of image data sets where usually images are not labeled.

4.Reinforcement Machine learning:

Reinforcement Machine Learning method that interacts with the environment by producing actions and discovering error. Trial, error, and delay are the most relevant characteristics of reinforcement learning. In this technique, the model keeps on increasing its performance using Reward Feedback to learn the behavior or pattern.

Random Forest Algorithms:

Random Forest algorithm is a powerful tree learning technique in Machine Learning to make predictions and **then we do voting of all the trees to make prediction**. They are widely used for classification and regression task.

- It is a type of classifier that uses many decision trees to make predictions.
- It takes different random parts of the dataset to train each tree and then it combines the results by averaging them;. This approach helps improve the accuracy of predictions. Random Forest is based on ensemble learning.

This study primarily concentrates on the Random Forest which is applied to forecast the yield of crops, which aids farmers in choosing and cultivating the most lucrative crop, so lowering the likelihood of loss and raising productivity and the value of farming land. Hyper parameter tuning is a key component of machine learning algorithms. It improves a machine learning model's effectiveness and is fine-tuned before the training process begins. Therefore, in this study, the Random Search algorithm has been used to optimize the Random Forest method's hyper parameters to improve the performance of a predictive model.

Benefits and Challenges of Random Forest

One advantage of the random forest algorithm is its versatility. This approach is applicable to both regression and classification issues. The algorithm might be deemed advantageous as it yields superior outcomes even in the absence of hyperparameter adjustment. Furthermore, they possess a high level of clarity, making them easily comprehensible. Additionally, their number is quite limited.

Overfitting is a significant issue in machine learning. We must develop a universal model capable of achieving satisfactory outcomes on the test data. The random forest overcomes this dilemma by aggregating multiple decision trees, resulting in reduced bias and volatility.

2.4 OBJECTIVES

- The main objective of crop yield prediction is to improve decision-making in agriculture, optimize resource use, mitigate risks, and ultimately enhance food security and sustainability by accurately forecasting potential yields.
- The Objective of the study is to provide a solution for farmers decide which crop to grow in addition to increasing yield.
- Farmers can benefit from yield estimation because it allows them to minimize crop loss and obtain the best prices for their crops.
- Crop yield predictions are carried out to estimate higher crop yield through the use of machine learning algorithms which are one of the challenging issues in the agricultural sector.

2.5 PROBLEM STATEMENT

Agriculture is one of the main sources of income in India. There is need to improve the sustainability of agriculture with the rate of increase in suicides of farmer due to crop failure and less yield and losses. Hence, it is a significant contribution towards the economic and agricultural welfare of the countries across the world. The Problem Statement revolves around prediction of yield of crops considering different climatic conditions of India including various attributes. Goal of this project is to help the farmers to choose the suitable crop to grow in order to get the required yield and the profit. Need for the crop yield prediction is very much essential at this point of time for selecting the right crop.

The problem statement for crop yield prediction using machine learning aims to develop a model that accurately forecasts crop yields based on various factors like weather patterns, soil conditions, and agricultural practices, ultimately empowering farmers and stakeholders with data-driven insights for improved decision-making.

3.LITERATURE REVIEW

3.LITERATURE REVIEW

K. G. Liakos et al., (2018), Machine learning has emerged with big data technologies and high-performance computing to create new opportunities for data intensive science in the multi-disciplinary agri-technologies domain. In this paper, we present a comprehensive review of research dedicated to applications of machine learning in agricultural production systems. The works analyzed were categorized in (a) crop management, including applications on yield prediction, disease detection, weed detection crop quality, and species recognition; (b) livestock management, including applications on animal welfare and livestock production; (c) water management; and (d) soil management. The filtering and classification of the presented articles demonstrate how agriculture will benefit from machine learning technologies. By applying machine learning to sensor data, farm management systems are evolving into real time artificial intelligence enabled programs that provide rich recommendations and insights for farmer decision support and action.

Kumar et al., (2018) concluded this works helps in improving the success of yield rate of crops by applying different classification methods by comparing the different parameters. Various machine learning algorithms compared to predict crop yield. The algorithms included for comparative analysis are Artificial neural network, support vector machine, K-Nearest Neighbors, Decision Tree, Random forest, Gradient boosted decision tree, regularized greedy Forest and proposed CSM technique (Crop Selection Method) which helps in predicting the sequence of crops can be considered for planning in the coming seasons. Author concluded that the prediction is based on many parameters so accuracy and performance of the system varies as per parameters.

Zhang et al. (2023) investigate the use of the Random Forest algorithm to improve crop yield prediction accuracy. By constructing an ensemble of decision trees and aggregating their predictions, Random Forest captures complex nonlinear relationships between predictor variables and crop yields. This research illustrates the superior performance of Random Forest over traditional regression models, making it an asset for precision agriculture and informed decision making in farming practices.

Ravisha & Neelam Sinha (2023), This study presents a robust and effective crop recommendation system, designed to be userfriendly for farmers across India. The system aids farmers in making data-driven decisions on the type of crop to cultivate based on several factors such as Humidity, Nitrogen, Phosphorus, pH Value, Potassium, Rainfall, and Temperature. Utilizing this research could potentially enhance the country's productivity and profitability. It empowers farmers to select the most suitable crop, thereby maximizing their yield and contributing to the nation's overall economic gain. The study explores the application of various machine learning algorithms, including Decision Tree, Naïve Bayes, Support Vector Machine, Logistic Regression, Random Forest, and XGBoost, for crop recommendation. An analysis of these six algorithms revealed that XGBoost delivered the most accurate results, with an accuracy score of 99.31%.

Payal Gulati & Suman Kumar Jha(2020) In this paper, various techniques of Machine learning have been executed on the agricultural data to evaluate the best performing technique. We utilize six different supervised learning algorithms. This proposed dataset comprises of a variety of parameters that are useful for identifying status of crop and leading training on datasets collected. This paper also shows the correlation of all six techniques. The results of these techniques were compared based on different errors and cross validation is to be done for obtaining accuracy. Here, Gradient Boosting Regressor is giving more accuracy with cross validation runs as 87.9% when target variable is "Yield" but when target variable is "Production", the Random Forest Regressor is providing more cross validation accuracy of 98.9%. This framework will assist to reduce the issues faced by farmers and will serve as delegate to provide farmers with the information they need to gain high and maximize the profits.

Khaki, S., & Wang, L. (2019), the Paper presented a machine learning approach for crop yield prediction, which demonstrated superior performance in the 2018 Syngenta Crop Challenge using large datasets of corn hybrids. The approach used deep neural networks to make yield predictions (including yield, check yield, and yield difference) based on genotype and environment data. The carefully designed deep neural networks were able to learn nonlinear and complex relationships between genes, environmental conditions, as well as their interactions from historical data and make reasonably accurate predictions of yields for new hybrids planted in new locations with known weather conditions. Performance of the model was found to be relatively sensitive to the quality of weather prediction, which suggested the importance of weather prediction techniques.

Padma T and Dipali Sinha(2023), In this investigation, the author employed various models of regression to forecast crop yield and observed that the random search optimized random forest technique performed better. According to results from experiments on crop yield data, the best outcome, with an R2 value of 0.98, is produced by an improved Random Forest. Despite the high forecast accuracy, this work still has certain shortcomings. Future research will examine more deep learning and machine learning models to create more accurate prediction systems.

Thomas van Klompenburg et.al (2020),This study showed that the selected publications use a variety of features, depending on the scope of the research and the availability of data. Every paper investigates yield prediction with machine learning but differs from the features. The studies also differ in scale, geological position, and crop. The choice of features is dependent on the availability of the dataset and the aim of the research. Studies also stated that models with more features did not always provide the best performance for the yield prediction. To find the best performing model, models with more and fewer features should be tested. Many algorithms have been used in different studies. The results show that no specific conclusion can be drawn as to what the best model is, but they clearly show that some machine learning models are used more than the others. The most used models are the random forest, neural networks, linear regression, and gradient boosting tree. Most of the studies used a variety of machine learning models to test which model had the best prediction.

4. SYSTEM PROPOSAL

4. SYSTEM PROPOSAL

4.1 EXISTING SYSTEM

- In our research, which we found in the previous research papers is that everyone uses climatic factors like rainfall, sunlight and agricultural factors like soil type, nutrients possessed by the soil (Nitrogen, Potassium, etc.)
- But the problem is we need to gather the data and then a third party does this prediction and then it is explained to the farmer and this takes a lot of effort for the farmer and he doesn't understand the science behind these factors.

DISADVANTAGES:

- **Limited Data Availability** – Many systems rely on incomplete or outdated datasets, reducing prediction accuracy.
- **Lack of Real-Time Updates** – Traditional models do not incorporate real-time weather, soil, or market conditions.
- **Low Precision in Small Regions** – Many models work well for large-scale predictions but struggle with localized variations.
- **Limited Use of Advanced ML Techniques** – Some existing systems use basic statistical methods instead of advanced machine learning models, leading to lower accuracy.
- **High Computational Costs** – More accurate models require significant computational power, which can be a challenge for small farmers.
- **Dependency on Historical Data** – Systems that rely solely on past data struggle to adapt to climate change and new farming techniques.
- **Poor Farmer Accessibility** – Many tools are complex, making them difficult for farmers with limited technical knowledge to use.

4.2 PROPOSED SYSTEM

The Proposed model is to predict the crop yield. We have collected the dataset for the crop prediction from real world sensor data available in kaggle.com. This dataset contains attributes such as soil, temperature, pH, Rainfall, Humidity.

For getting high accuracy we used the Random Forest algorithm which gives accuracy which predicate by model and actual outcome of predication in the dataset. In crop yield prediction, "random classification" refers to using machine learning classification algorithms, like Random Forest, to categorize crops or predict yield based on factors like weather, soil, and crop characteristics, rather than predicting a continuous yield value.

Simplified Prediction:

Instead of trying to predict a precise yield value, which can be complex, classification simplifies the task by focusing on yield categories.

Decision Support:

Farmers can use these models to make decisions about which crops to plant, when to harvest, and what resources to allocate based on predicted yield levels.

Early Warning Systems:

Classification models can help identify potential yield problems early on, allowing farmers to take corrective actions.

STEPS

- Define a problem
- Preparing data
- Evaluating algorithms
- Improving results
- Predicting results

ADVANTAGES:

- **Improved Decision-Making**

Helps farmers and agricultural planners make informed choices about planting, irrigation, and harvesting.

- **Resource Optimization**

Efficient use of water, fertilizers, and pesticides based on predicted yields, reducing waste and costs.

- **Increased Productivity**

Enables better planning, leading to higher crop yields and profitability.

- **Climate Adaptation**

Assists farmers in adapting to changing weather conditions by analyzing historical climate data.

- **Market and Price Forecasting**

Assists farmers in making informed selling decisions by predicting market supply and demand trends.

- **Sustainability**

Promotes eco-friendly farming by minimizing resource wastage and environmental impact.

5.SYSTEM REQUIREMENTS

5. SYSTEM REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

Processor : Any Processor Above 1 GHz

RAM : Minimum 4 GB

Hard Disk : Few (5 GB) free Disk Space

Input Device : Standard Keyboard and Mouse

Output Device : A monitor with 1024x768 resolution is sufficient

5.2 SOFTWARE REQUIREMENTS

Operating System : Windows 7 or Higher

Programming : Python is used within Google colab

5.3 SOFTWARE DESCRIPTION

Google Colab, short for Collaboratory, is a free cloud-based platform provided by Google that allows users to write and execute Python code collaboratively in a Jupyter Notebook environment. Google Collaboratory notebook, is designed to facilitate machine learning (ML) and data science tasks by providing a virtual environment, Google colab python with access to free GPU resources. Colab offers free GPU access, which is particularly useful for training machine learning models that require significant computational power.

Colab runs in the cloud, eliminating the need for users to set up and configure their own development environment. This makes it convenient for quick coding and collaboration. Colab is integrated with Google Drive, allowing users to save their work directly to their Google Drive account. This enables easy sharing and access to notebooks from different devices.

Colab comes pre-installed with many popular Python libraries for machine learning, data analysis, and visualization, such as TensorFlow, PyTorch, Matplotlib, and more.

Colab notebooks can be easily shared just like Google Docs or Sheets. Users can provide a link to the notebook, and others can view or edit the code in real-time.

Python is one of the most popular programming languages for machine learning, both in academia and in industry. As such, it is essential to learn this language for anyone interested in machine learning. Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python consistently ranks as one of the most popular programming languages, and has gained widespread use in the machine learning community.

Google Colab, short for Google Colaboratory, is a free cloud service provided by Google that allows users to run Python code real-time in a browser-based environment. It offers a convenient way to write, execute, and share Python code along with its output. Colab is a hosted Jupyter Notebook service that requires no setup to use and provides free of charge access to computing resources, including GPUs and TPUs. Colab is especially well suited to machine learning, data science, and education. Yes. Colab is free of charge to use.

6. SYSTEM DIAGRAM

6. SYSTEM DIAGRAM

6.1 SYSTEM ARCHITECTURE

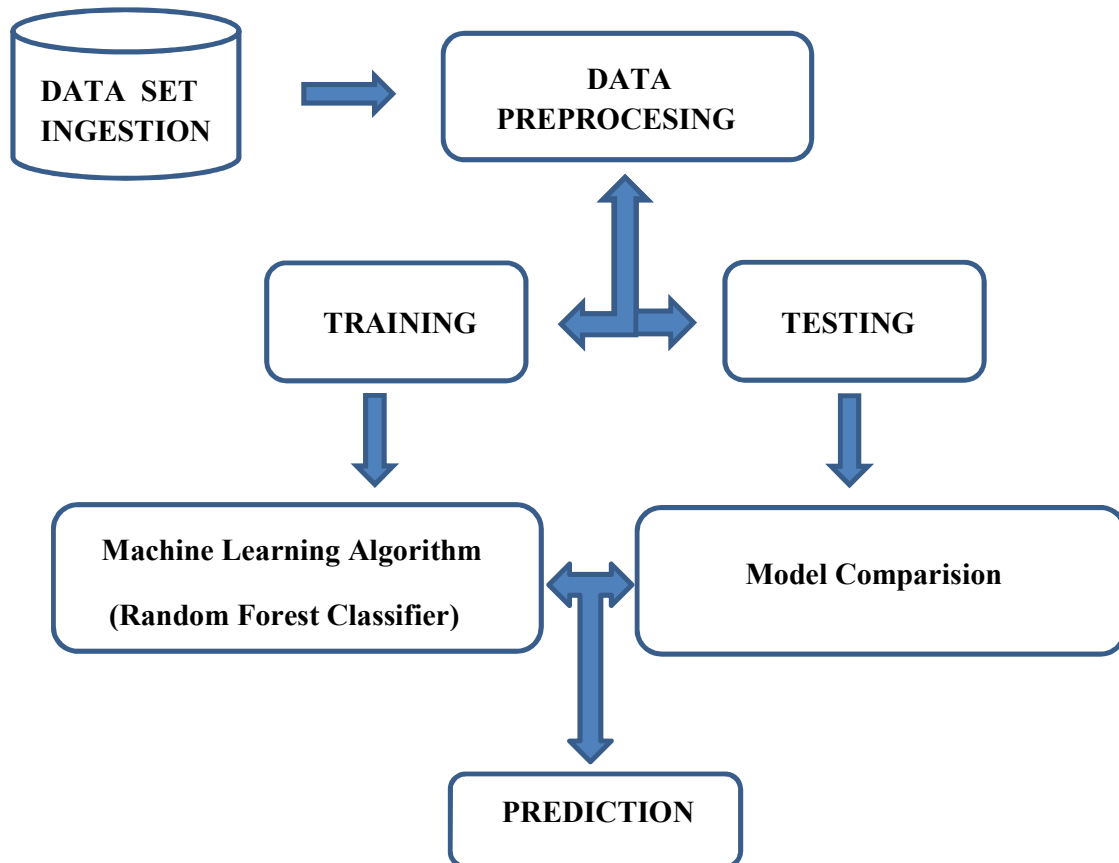


Fig.2: System Architecture for the Proposed Model

7. IMPLEMENTATION

7. IMPLEMENTATION

7.1 MODULE LIST

The “CROP YIELD PREDICTION USING MACHINE LEARNING” module is a core component of an agriculture decision support system. It uses Machine Learning (ML) and data analytics to predict crop yields based on various factors, including weather patterns, soil properties, and farming practices. The Module List contains,

- Data Preprocessing
- Data Loading
- Data exploration
- Data Analysis through Visualization
- Prediction

7.2 MODULE DESCRIPTION

7.2.1: DATA PREPROCESSING:

Preprocessing the data is considered as a significant step machine learning phase. Preprocessing involves adding the missing values, the correct set of data, and extracting the functionality. Data set form is important to the process of analysis. The data collected in this step will be induced in Google Colab platform in the form of python programming in order to get the desired output.

7.2.2: DATA LOADING:

Python script code that is being executed in Google Colab, used to load and display a CSV dataset. Pandas is used for data manipulation and analysis, particularly working with data in tabular form (e.g., CSV files). Used to mount Google Drive in the Colab environment, allowing access to files stored in your Google Drive. Shape of the DataFrame should display the number of rows and columns in the dataset using, the dataset contains 2200 rows and 8 columns.

The code runs without errors, indicating that the CSV file was successfully loaded into a pandas DataFrame. the first few rows of the dataset displayed, confirming that the `display(df.head())` command is working properly. The dataset appears to contain the following columns:

N: Nitrogen content in the soil.

P: Phosphorus content in the soil.

K: Potassium content in the soil.

temperature: The temperature in degrees Celsius.

humidity: The humidity percentage.

ph: The pH level of the soil.

rainfall: The amount of rainfall in mm.

label: The recommended crop

The script properly mounts Google Drive, uploads the CSV file, and reads it into a pandas DataFrame. The dataset contains soil and weather parameters along with the recommended crop. The script handles file loading errors with try-except blocks, ensuring smooth execution even if the file is missing or there are other issues.

	N	P	K	temperature	humidity	ph	rainfall	label
0	90	42	43	20.879744	82.002744	6.502985	202.935536	rice
1	85	58	41	21.770462	80.319644	7.038096	226.655537	rice
2	60	55	44	23.004459	82.320763	7.840207	263.964248	rice
3	74	35	40	26.491096	80.158363	6.980401	242.864034	rice
4	78	42	42	20.130175	81.604873	7.628473	262.717340	rice

Shape of the DataFrame:
(2200, 8)

Fig.3: Output of the Given Data Frame

7.2.3: DATA EXPLORATION:

Python code snippet used for data exploration in a data analysis or machine learning project. This code performs a basic data exploration process by:

- Checking the shape and types of the dataset.

(Number of rows and columns, the dataset contains both integer and float values.)

- Identifying missing values.

(Checks for missing (null) values in each column.)

- Summarizing numerical features.

(List of numerical columns. Provides summary statistics such as mean, standard deviation, min, max, and quartiles for the numerical columns)

- Analyzing unique values and their frequencies for categorical features.

(List of categorical columns. Displays all unique values in the categorical column.

Shows the frequency of each unique value).

```
Data Types:
N          int64
P          int64
K          int64
temperature float64
humidity   float64
ph         float64
rainfall   float64
label      object
dtype: object
```

```
Shape: (2200, 8)
```

```
Missing Values:
N          0
P          0
K          0
temperature 0
humidity    0
ph          0
rainfall    0
label       0
dtype: int64
```

```
Descriptive Statistics for Numerical Features:
```

	N	P	K	temperature	humidity
\					
count	2200.000000	2200.000000	2200.000000	2200.000000	2200.000000
mean	50.551818	53.362727	48.149091	25.616244	71.481779
std	36.917334	32.985883	50.647931	5.063749	22.263812
min	0.000000	5.000000	5.000000	8.825675	14.258040
25%	21.000000	28.000000	20.000000	22.769375	60.261953
50%	37.000000	51.000000	32.000000	25.598693	80.473146
75%	84.250000	68.000000	49.000000	28.561654	89.948771
max	140.000000	145.000000	205.000000	43.675493	99.981876

	ph	rainfall
count	2200.000000	2200.000000
mean	6.469480	103.463655
std	0.773938	54.958389
min	3.504752	20.211267
25%	5.971693	64.551686
50%	6.425045	94.867624
75%	6.923643	124.267508
max	9.935091	298.560117

```

Unique values in 'label':
['rice' 'maize' 'chickpea' 'kidneybeans' 'pigeonpeas' 'mothbeans'
 'mungbean' 'blackgram' 'lentil' 'pomegranate' 'banana' 'mango'
 'grapes'
 'watermelon' 'muskmelon' 'apple' 'orange' 'papaya' 'coconut' 'cotton'
 'jute' 'coffee']

Frequency of each category in 'label':
label
rice          100
maize         100
chickpea      100
kidneybeans   100
pigeonpeas    100
mothbeans     100
mungbean      100
blackgram     100
lentil        100
pomegranate   100
banana        100
mango         100
grapes        100
watermelon    100
muskmelon     100
apple         100
orange        100
papaya        100
coconut       100
cotton        100
jute          100
coffee       100
Name: count, dtype: int64

```

Fig.4: Data Exploration output

7.2.4: DATA ANALYSIS THROUGH VISUALIZATION:

We import **Matplotlib**: **Matplotlib** is a popular data visualization library in python. It provide a wide variety of plotting functions to create static, animated, and interactive visualization. It is wiedy used for creating line plot, barchars, scatter plots, histograms, pie charts, and more.

Pandas:

The Pandas library in Python is wiedy used for data manipulation and analysis. It provides data structure and functions to efficiently handle and process large dataset.

Seaborn:

Seaborn is a library that provides high-level, attractive, and customizable plots for statistical data analysis. Seaborn makes it easier to generate complex plot with minimal code, offering better aesthetics and build-in support dor data frame. We use **pd.read_csv()** to load the dataset.

1. Correlation Matrix:

The correlation matrix of numerical features, visualized using a heatmap. It displays the correlation coefficients between pairs of variables.

Correlation values range from -1 to 1:

1.00 → Perfect positive correlation (red)

-1.00 → Perfect negative correlation (blue)

0.00 → No correlation

The diagonal values are all 1.00 since they represent the correlation of each variable with itself.

The variables in this matrix are N (Nitrogen), P (Phosphorus), K (Potassium), Temperature, Humidity, pH, Rainfall.

Observations

- N and P: Correlation of -0.23 → Slight negative relationship.
- P and K: Correlation of 0.74 → Strong positive correlation.
- Temperature and humidity: Correlation of 0.21 → Weak positive relationship.
- pH and rainfall: Correlation of -0.11 → Slight negative correlation.
- K and temperature: Correlation of 0.16 → Weak positive relationship.
- Most of the correlations are weak or moderate, with P and K having the strongest positive correlation.

Color Interpretation

Red shades → Positive correlations.

Blue shades → Negative correlations.

The intensity of the color indicates the strength of the correlation.

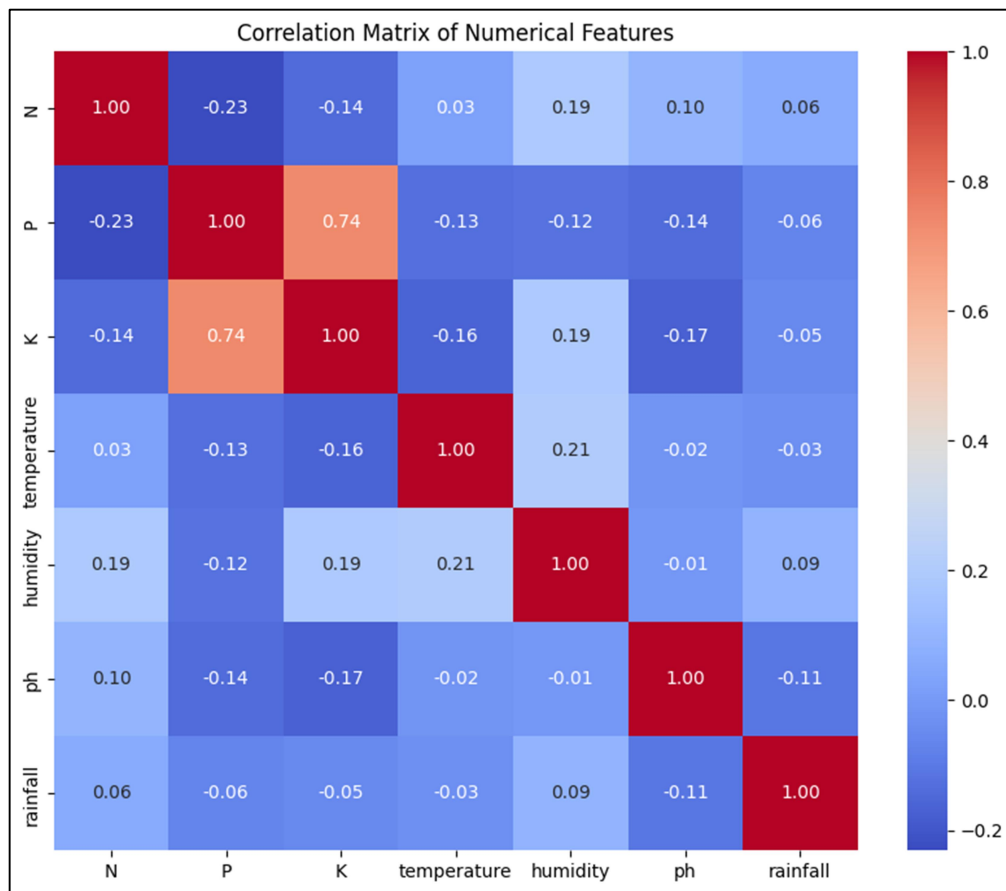


Fig.5: Correlation Matrix of Numerical features

1. Box Plots:

These box plots compare different variables against various crop types. Multiple Variables Analyzed.

Box Plot Characteristics:

Each box plot displays the distribution of a variable (e.g., pH, humidity) across different crop types.

The boxes represent the interquartile range (IQR), while the whiskers show variability outside the upper and lower quartiles.

Purpose:

The visual likely aims to show how different environmental or soil factors vary across different crops. This could be used for agricultural analysis, crop yield prediction, or soil condition assessment.

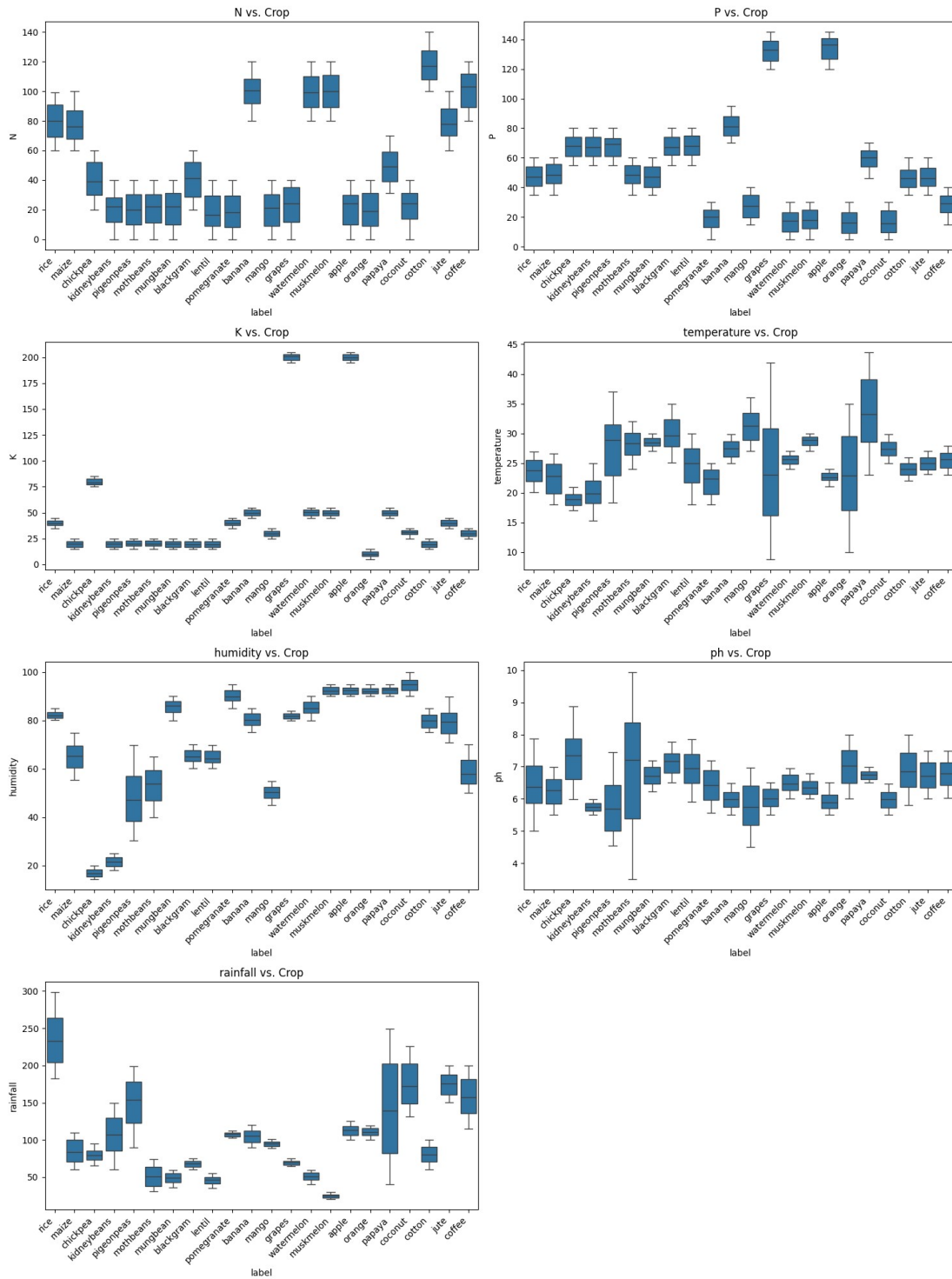


Fig.6: Box plots of numerical features for each crop type

2. Scatter Plots:

A scatter plot is a type of graph used to visualize the relationship between two numerical variables. It uses dots (or points) to represent individual data points. The position of each dot on the graph is determined by the values of the two variables being examined. Typically, one variable is plotted on the horizontal axis (x-axis), and the other on the vertical axis (y-axis).

Positive Correlation:

Picture a line sloping upwards from the bottom left to the top right. If the dots generally follow this pattern, there's a positive correlation.

Negative Correlation:

Positive Correlation: Picture a line sloping upwards from the bottom left to the top right. If the dots generally follow this pattern, there's a positive correlation. This means as the x-variable increases, the y-variable tends to increase too.

No Correlation:

If the dots are scattered randomly, like confetti, there's no clear correlation. The variables don't seem to have any relationship.

Purpose:

The goal is to identify the relationship between key agricultural parameters (nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall) and specific crop types. This helps in understanding the ideal conditions for growing different crops, which is essential for your crop recommendation .

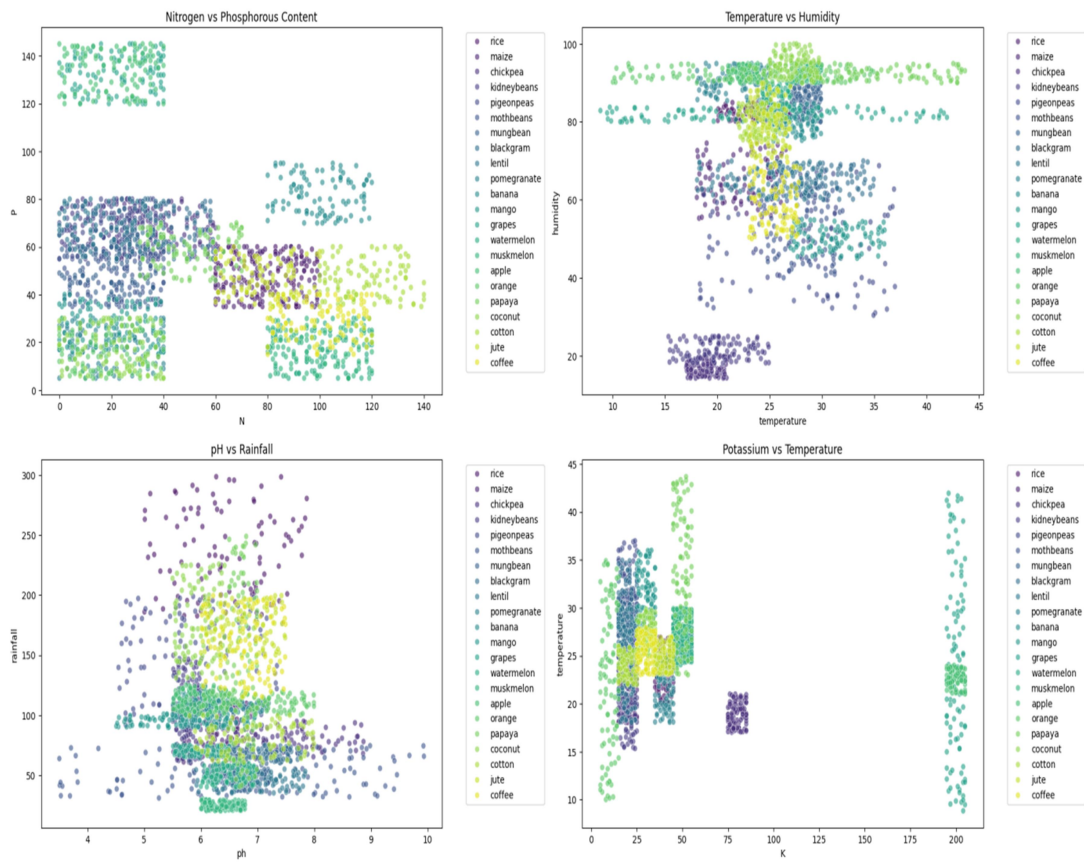


Fig.7: Scatter Plot

3. Mean Values:

Group the data by crop type (label) and calculate the mean for each numerical feature. This summarizes the average conditions preferred by each crop. These means provide a crop, highlighting typical nutrient levels, pH, and rainfall conditions.

4. Comprehensive EDA:

The code effectively summarizes and visualizes crop data, making it easier to identify patterns, and crop-specific characteristics.

7.2.5: PREDICTION:

1. Import libraries:

sklearn (scikit-learn) is a very popular and powerful python library used for machine learning. It provides a wide range of tools for machine learning and statistical modeling. It's simple and efficient tools for data preprocessing, model selection supervised and unsupervised learning algorithms. More using numpy, pandas, matplotlib, seaborn import libraries.

2. GUI:

tkinter.messagebox Imported but not used in this code. It could be used to display message boxes in graphical user interface(GUI)

3. Input :

The code uses input() to collect soil and weather conditions from the user:

- n: Nitrogen content.
- P: Phosphorus content.
- k: Potassium content.
- t: Temperature
- h: Humidity
- ph: pH level of the soil.
- r: Rainfall

Each input is converted to float, int, to ensure the value are treated as numerical data the value are appended to the list features, which will be used later to make a prediction.

4. Model Training:

It trains a Random Forest Classifier on the crop dataset.

5. Evaluation:

The program evaluates the model using multiple metrics.

Accuracy: Overall correctness of the model.

Precision: Correct positive prediction out of total predicted positives.

Recall: Correctly identified positive cases.

F1 Score: The Harmonic mean of precision and recall.

6. Storing and displaying Metrics:

Create a list classifiers containing the random_forest model. For each model function evaluate() is called to calculate the performance metrics. The metrics are converted into a Data frame. The data Frame is appended to list dfs. Finally, all DataFrame are concatenated into results_df.

7. Prediction:

Predict Probabilities for each crop based on the user input features. Displaying prediction probabilities to create a data Frame to store the prediction probabilities. The Data frame is melted into long format for plotting.

8. Visualization:

display the predicted probabilities using a bar plot.

9. Recommendation:

The crop with the highest probability and display it as the recommendation.

```

importNitrogen45
Phosphorus67
Potassium23
Temperature12
Humidity56
pH8
Rainfall56
Accuracy: 0.99318181818182
Precision: 0.9937348484848485
Recall: 0.99318181818182
F1 Score: 0.9931754816901672
Accuracy: 0.99318181818182
Precision: 0.9937348484848485
Recall: 0.99318181818182
F1 Score: 0.9931754816901672

```

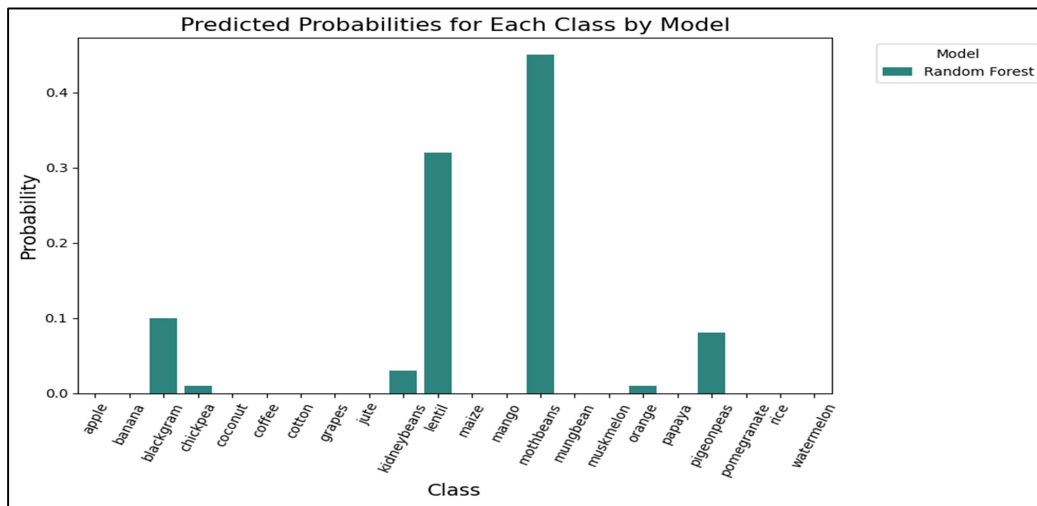


Fig.8: Prediction Output

8. CONCLUSION

8. CONCLUSION

Crop yield prediction is crucial for the farmer to make decisions. Numerous studies have been conducted in the agricultural field to forecast crop production. We observed that the random forest technique performed better. According to results from experiments on crop yield data, the best outcome, with an accuracy value of 0.99, is produced by an improved Random Forest. Despite the high yield accuracy, this work still has certain shortcoming. The use of machine learning, particularly the Random Forest algorithm, is becoming increasingly prevalent in crop yield prediction. This involves forecasting the amount of crops that will be produced, which is crucial for agricultural planning, resource management, and food security.

Machine learning method that combines multiple decision trees to improve prediction accuracy. It's effective in handling complex relationships between input features and crop yields. It is robust against overfitting, which is a common problem in machine learning.

Analyzing various environmental and agricultural factors, machine learning models can provide valuable insights for farmers and agricultural stakeholders. Accurate yield predictions can help optimize resource allocation, improve farming practices, and enhance food security. The use of machine learning in agriculture is helping to revolutionize farming practices. In essence, the application of Random Forest algorithms within crop yield prediction, is providing a very valuable tool for those working within the agricultural sector.

9. FUTURE ENHANCEMENT FOR PHASE II

9. FUTURE ENHANCEMENT FOR PHASE II

Future enhancements for crop yield prediction using machine learning include leveraging advanced algorithms like deep learning (CNN, LSTM, DNN), incorporating more data sources (remote sensing, IoT), and focusing on real-time prediction and personalized recommendations for farmers.

Implement real-time prediction systems that can provide timely insights and alerts to farmers. Develop regional and crop-specific models to account for variations in climate, soil, and agricultural practices.

Incorporate XAI techniques to provide insights into the model's predictions, helping to build trust and understanding among users. Implement robust model monitoring systems to track performance and identify potential issues. Establish feedback loops to collect data from farmers and agricultural experts to improve the model's accuracy and relevance.

APPENDICES

APPENDICES

DATA LOADING:

Load the dataset "Crop_recommendation.csv" into a pandas Data Frame.

```
import pandas as pd
from google.colab import drive
from google.colab import files
drive.mount('/content/drive')
uploaded = files.upload()

try:
    df = pd.read_csv('/content/Crop_recommendation.csv')
    display(df.head())
    print()
    print("Shape of the DataFrame:")
    print(df.shape)
except FileNotFoundError:
    print("Error: 'Crop_recommendation.csv' not found.")
    df = None
except Exception as e:
    print(f"An error occurred: {e}")
    df = None
```

DATA EXPLORATION:

Explore the loaded dataset to understand its characteristics.

```
# Data Types and Shape
print("Data Types:\n", df.dtypes)
print("\nShape:", df.shape)

# Missing Values
print("\nMissing Values:\n", df.isnull().sum())

# Descriptive Statistics for Numerical Features
numerical_features = ['N', 'P', 'K', 'temperature', 'humidity',
                      'ph', 'rainfall']
print("\nDescriptive Statistics for Numerical Features:\n"
      df[numerical_features].describe())

# Unique Values and Frequencies for Categorical Features
categorical_features = ['label']
for col in categorical_features:
    print(f"\nUnique values in '{col}':\n{df[col].unique()}")
    print(f"\nFrequency of each category in"
          '{col}':\n{df[col].value_counts()}")
```

DATA ANALYSIS:

Analyze the relationships between different features in the dataset, with a focus on the relationship between input features and the 'label' (crop).

```
import matplotlib.pyplot as plt
import seaborn as sns
import seaborn as sns

df = pd.read_csv('/content/Crop_recommendation.csv')

# Calculate the correlation matrix
correlation_matrix = df.corr(numeric_only=True)

# Visualize the correlation matrix using a heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix of Numerical Features')
plt.show()

# Box plots of numerical features for each crop type
numerical_features = ['N', 'P', 'K', 'temperature', 'humidity', 'ph',
                      'rainfall']
plt.figure(figsize=(15, 20))
for i, feature in enumerate(numerical_features):
    plt.subplot(4, 2, i + 1)
    sns.boxplot(x='label', y=feature, data=df)
    plt.title(f'{feature} vs. Crop')
    plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

plt.figure(figsize=(20, 12))
```

```

plt.subplot(2, 2, 1)
sns.scatterplot(data=df, x='N', y='P', hue='label', palette='viridis',
alpha=0.6)
plt.title('Nitrogen vs Phosphorous Content')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

# Temperature vs Humidity
plt.subplot(2, 2, 2)
sns.scatterplot(data=df, x='temperature', y='humidity', hue='label',
palette='viridis', alpha=0.6)
plt.title('Temperature vs Humidity')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

# pH vs Rainfall
plt.subplot(2, 2, 3)
sns.scatterplot(data=df, x='ph', y='rainfall', hue='label',
palette='viridis', alpha=0.6)
plt.title('pH vs Rainfall')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

# K vs Temperature
plt.subplot(2, 2, 4)
sns.scatterplot(data=df, x='K', y='temperature', hue='label',
palette='viridis', alpha=0.6)
plt.title('Potassium vs Temperature')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

# Calculate the mean value of each numerical feature for each crop
crop_means = df.groupby('label')[numerical_features].mean()

```

```
display(crop_means)

# Summarize observations

print("Observations:")

print("The correlation matrix heatmap shows the linear
relationships between numerical features. The box plots visualize the
distribution of each numerical feature for each crop, revealing how
they relate to different crop types. Highlighting potential differences
in optimal growing conditions. The mean values provide a profile for
each crop.")
```

PREDICTION:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score, confusion_matrix

from tkinter import messagebox

n=float(input("Nitrogen"))
p=float(input("Phosphorus"))
k=float(input("Potassium"))
t=float(input("Temperature"))
h=float(input("Humidity"))
ph=float(input("pH"))
r=float(input("Rainfall"))

features=[]
features.append(n)
features.append(p)
features.append(k)
features.append(t)
features.append(h)
features.append(ph)
features.append(r)

df = pd.read_csv('/content/Crop_recommendation.csv')

#Train Machine Learning Models

X = df.drop('label', axis=1)
y = df['label']
class_names = np.unique(y)
```

```

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

# classifiers
random_forest = RandomForestClassifier(n_estimators=100,
random_state=42)

# Train models
random_forest.fit(X_train, y_train)

def evaluate(model, conf_matrix=True, return_=False):

    # Confusion Matrix
    y_pred = model.predict(X_test)
    cm = confusion_matrix(y_test, y_pred)

    # Calculating and displaying additional metrics
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, average='weighted')
    recall = recall_score(y_test, y_pred, average='weighted')
    f1 = f1_score(y_test, y_pred, average='weighted')
    print(f"Accuracy: {accuracy}")
    print(f"Precision: {precision}")
    print(f"Recall: {recall}")
    print(f"F1 Score: {f1}")

    # Extracting the model name for display
    model_name = str(type(model).__name__)

    # Returning metrics as a dictionary if specified

```

```

        if return_:
            metrics_dict = {
                "Model": model_name,
                "accuracy": round(accuracy, 4),
                "precision": round(precision, 4),
                "recall": round(recall, 4),
                "f1": round(f1, 4)
            }

            return metrics_dict
evaluate(random_forest)

#capture
# List of classifiers
classifiers = [
    random_forest
]

# Initialize an empty list to store DataFrames
dfs = []

# Loop through classifiers
for classifier in classifiers:
    # Calculate metrics for the current classifier
    metrics_result = evaluate(classifier, return_=True,
conf_matrix=False)

    # Create a DataFrame from the dictionary
    metrics_df = pd.DataFrame([metrics_result])

    # Append the DataFrame to the list
    dfs.append(metrics_df)

# Concatenate DataFrames into a single DataFrame
results_df = pd.concat(dfs, ignore_index=True)

# Format the metrics to 4 decimal places in the DataFrame

```

```

results_df["accuracy"] = results_df["accuracy"].map("{:.4f}".format)
results_df["precision"] = results_df["precision"].map("{:.4f}".format)
results_df["recall"] = results_df["recall"].map("{:.4f}".format)
results_df["f1"] = results_df["f1"].map("{:.4f}".format)

results_df

#Comparison of Models' Performance Metrics

results_df["precision"] = results_df["precision"].astype(float)
results_df["recall"] = results_df["recall"].astype(float)
results_df["f1"] = results_df["f1"].astype(float)

#Make a Single Prediction Using Trained Models

# Predict probabilities for each class
rf_probs = random_forest.predict_proba([features])[0]

# Create a DataFrame to store the probabilities
probs_df = pd.DataFrame({
    'Class': class_names,
    'Random Forest': rf_probs,

    })

# Melt the DataFrame for Seaborn
probs_df_melted = probs_df.melt(id_vars='Class', var_name='Model',
value_name='Probability')

# Plot using Seaborn
plt.figure(figsize=(10, 6))

```

```

sns.barplot(x='Class', y='Probability', hue='Model',
data=probs_df_melted, palette='viridis')
plt.title('Predicted Probabilities for Each Class by Model',
fontsize=16)
plt.xlabel('Class', fontsize=14)
plt.ylabel('Probability', fontsize=14)
plt.xticks(rotation=65) # Rotate class names for better readability
plt.legend(title='Model', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
predicted_class = class_names[np.argmax(rf_probs)]
print(f"\nRecommended crop is: {predicted_class}")

```

SCREENSHOT AND SAMPLE OUTPUT

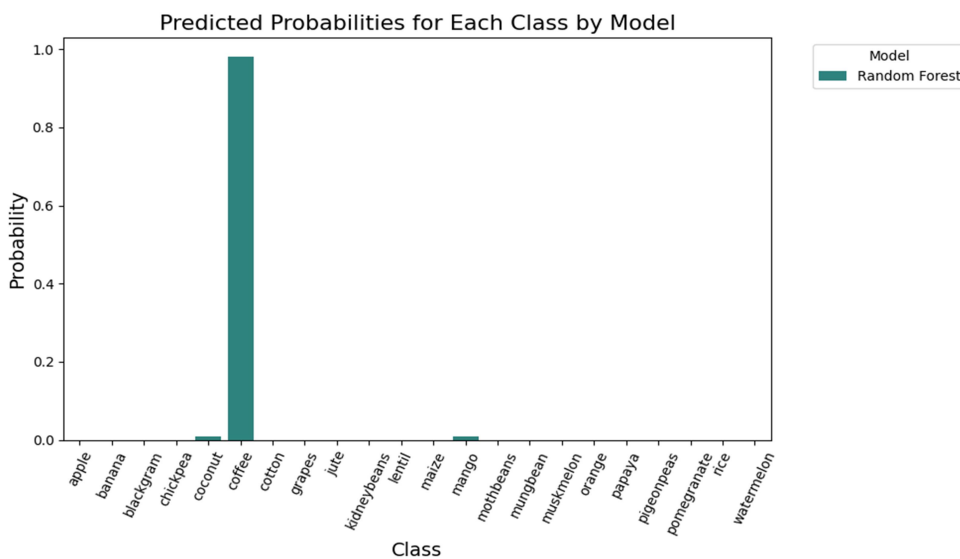
SAMPLE OUTPUT

Sample Input:

Nitrogen99
Phosphorus15
Potassium27
Temperature27.42
Humidity56.64
pH6.09
Rainfall127.92

Output:

Accuracy: 0.9931818181818182
Precision: 0.9937348484848485
Recall: 0.9931818181818182
F1 Score: 0.9931754816901672
Accuracy: 0.9931818181818182
Precision: 0.9937348484848485
Recall: 0.9931818181818182
F1 Score: 0.9931754816901672



Recommended crop is: coffee

Sample Output of the Predicted Model

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REFERENCES

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